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Classifier Write-up

Best Accuracy Achieved on Test Set

0.80, using Method 3: Instance Weighting Based on Annotator Agreement

1. **Majority Vote Aggregation**

   For each review, I considered the label chosen by the majority of annotators as the "true" label. Using an Excel formula, I identified cases where the majority was greater than four and saved the results in a new column. The model performed above average, but false negatives were high. Some annotators may have been biased toward positive or negative classifications, but the majority vote treats all annotators equally, so occasional unreliable labels still influenced the outcome.

   Accuracy on the test set: 0.67

2. **Weighted Majority Vote by Annotator Reliability**

   I measured how often each annotator agreed with the majority vote to determine their reliability. Annotators with higher reliability were given more weight when deciding the final label for each review. I then trained a Naive Bayes classifier on this weighted dataset. Since some annotators were consistently more accurate, weighted voting helped reduce the influence of less reliable annotators.

   On the test set, this method achieved an accuracy of 0.70. Its performance was slightly similar to the simple majority vote, likely because the reliability differences between annotators were not very high.

3. **Instance Weighting Based on Annotator Agreement**

I measured the level of agreement among annotators for each review. Instances with higher agreement were given greater weight in the training data, while low-agreement instances were down-weighted. Weighted instances were replicated proportionally, and a Naive Bayes classifier was trained on the resulting dataset. This approach helps reduce noise from reviews where annotators disagreed substantially. Accuracy on the test set was 0.80. The instances in this set provided strong, reliable examples for the model to learn from, making this method the most effective of the three.